Settling the unconfirmed implications of the order-3 exploration: findings

23 October 2026

1 Starting point

The fourth exploration of the 149 representatives of the ETP laws of order at most three (`results/order3_vampire_mace4_etp_refutations_duals_asked`) left 93 implications *unconfirmed* (60 up to duality) and 5 *open questions*, with 1,652 laws in their conclusions that no prover had derived. Vampire, Prover9, Twee and Mace4 had failed on them with limits of up to a minute, and on single questions of up to 30 minutes. Unconfirmed implications steer the enumeration, so each false one makes the theory after its question unreliable.

Four ways of settling them were tried: a further prover (E), the log of the running exploration over all 1432 ETP equivalence classes, linear magmas, and translation-invariant magmas.

2 Summary of results

Question	Claim (non-trivial premise laws)	Refuted by	Status
82	Eq3, Eq75, Eq271 $\to x = y$ and 133 more laws	$x * y = 13x + 7y$ on $\mathbb{Z}/19$	checked on the table
135	dual of question 82	$x * y = 7x + 13y$ on $\mathbb{Z}/19$	checked on the table
171	Eq3, Eq49, Eq260 $\to$ Eq10, Eq378 and 16 more laws	translation-invariant magma of size 13	checked on the table
153	Eq3, Eq55, Eq105, Eq209, Eq264 $\to$ 60 laws	free magma, by E's saturation	not certified

Three unconfirmed implications (four with the dual) are thus false, three of them with finite counterexamples whose tables can be checked independently. The other 89 unconfirmed implications and the 5 open questions remain undecided. Since question 82 was accepted as unconfirmed and steered the enumeration from then on, the exploration has to be resumed at question 82.

3 The E prover

Each law of the conclusion of each unconfirmed implication and open question was posed to E 3.5.1 on its own, with the premise as axioms, a law and its dual under the dual premise being posed once: 1,087 problems. With 2 seconds and with 60 seconds per problem (`--auto-schedule`), the outcome was the same: no law was proved, E gave up on 847, and reported 240 as CounterSatisfiable. Of these, 180 belong to the open questions (with duals), where E agrees with Vampire.

The other 60 are the whole conclusion of question 153, whose premise

$$\text{Eq1}, \text{Eq3}, \text{Eq8}, \text{Eq23}, \text{Eq47}, \text{Eq55}, \text{Eq99}, \text{Eq105}, \text{Eq151},$$
$$\text{Eq203}, \text{Eq209}, \text{Eq255}, \text{Eq264}, \text{Eq307}, \text{Eq326}, \text{Eq359}, \text{Eq375}$$

amounts to idempotence ($x = x * x$) together with Eq55: $x = x * (y * (y * x))$, Eq105: $x = x * ((y * x) * y)$, Eq209: $x = (x * (y * x)) * y$ and Eq264: $x = ((x * y) * y) * y$. E saturates it within half a second for each law, with `--auto-schedule` and with `--auto`. If the saturation is correct, none of the 60 laws is derivable, and the free magma over the premise refutes the implication; Vampire, given one second per law of the free magma's row, had decided none of them.

Caveats. This is not a proof. E prints the saturated clause set but no certificate of its saturation, and the set contains equations that the term order cannot orient, such as $(x * y) * y = x * (y * x)$, so the Knuth–Bendix check of `completion.py`, which needs every rule to decrease, does not apply. The evidence is of the same kind as Vampire's saturation, which the exploration already trusts for its free magmas. No finite counterexample is known: an exhaustive Z3 search found no magma of size at most 7 that satisfies the premise and violates a law of the conclusion, and none of the searches below found one.

4 The exploration over all ETP equivalence classes

The log of the exploration over all 1432 ETP equivalence classes (Vampire expert, still running) was checked against the 98 questions: its 657 confirmed implications, the laws proved within its unconfirmed answers and the duals of both, closed under the ETP implication graph; its 654 finite counterexamples and their duals, evaluated on the order-3 laws; and its 14 free magmas and their duals, transferred through the ETP graph. None of the questions is settled, and the same closure reproves none of the 767 confirmed implications of the order-3 run. The reason is that the two explorations have not met: none of the 1,366 premises asked so far contains a law of order at most three besides $x = x$; they consist of laws numbered 4284 and above. In the lectic order, premises with these laws come late.

5 Linear magmas

A linear magma is $x * y = ax + by + c$ over $\mathbb{Z}/n$, or $x * y = Ax + By$ over $(\mathbb{Z}/p)^2$ with 2×2 matrices A and B. Every term is then a linear form in its variables, and a law holds exactly when its two sides have the same coefficients, so each magma is decided on all 149 laws without its table. All moduli $n < 130$ (with $c \in \{0, 1\}$) and all matrices over $\mathbb{Z}/2$ and $\mathbb{Z}/3$ were tried. They refute two questions:

- question 82, $\text{Eq3}, \text{Eq75}, \text{Eq271} \to x = y, \ldots$, with Eq75: $x = y * (y * (y * x))$ and Eq271: $x = ((y * x) * x) * y$, by $x * y = 13x + 7y$ on $\mathbb{Z}/19$, idempotent since $13 + 7 \equiv 1$; the implication claimed that the premise forces the trivial magma;
- question 135, its dual (asked separately because of `--ask-unconfirmed-duals`), by the dual magma $x * y = 7x + 13y$.

A magma of size 19 is beyond the reach of the expert's model search (Vampire for 10 seconds, Mace4 for 30 seconds per law).

6 Translation-invariant magmas

A translation-invariant magma is $x * y = x + f(y - x)$ on $\mathbb{Z}/n$; the idempotent linear magmas over $\mathbb{Z}/n$ are among them. Translations are automorphisms, so a law holds when it holds with its first variable 0, and the magma is determined by the n values of f instead of n^2 entries. A SAT encoding (CaDiCaL through `python-sat`) gives the functions f under which the premise holds; each is checked against the conclusion and excluded if it satisfies it, up to 200 per size and question. The encoding was checked against direct evaluation on 23,840 pairs of random functions and laws.

Up to size 21 for all 98 questions, this refutes question 171, $\mathrm{Eq3}, \mathrm{Eq49}, \mathrm{Eq260} \to \mathrm{Eq10}, \ldots$, with Eq49: $x = x * (x * (y * x))$ and Eq260: $x = ((x * y) * x) * x$, by

$$x * y = x + f(y - x) \bmod 13, \qquad f = (0, 2, 0, 4, 8, 6, 0, 8, 0, 9, 2, 2, 12),$$

which violates all 18 laws of its conclusion, among them Eq10: $x = x * (y * x)$ and Eq378: $x * y = (x * y) * y$. The premise is self-dual, so there is no dual question. It also confirms that question 82 has no translation-invariant counterexample smaller than 19. For some questions and sizes, the premise has more than 200 translation-invariant models, all satisfying the conclusion; there the search is not exhaustive. The search up to size 30 is still running.

7 Consequences and next steps

- Resume the fourth exploration at question 82, with the three refuting magmas in the context (`--context-magmas`); questions 135 and 171 are then refuted as well.
- Decide whether to accept question 153 on E's saturation, or to certify it first, by checking the ground confluence of E's saturated system or by describing the free magma explicitly.
- Further classes of magmas: linear magmas over finite fields $\mathbb{F}_{p^k}$ and with larger matrices; a conflict budget per SAT call to reach larger translation-invariant magmas.
- On the proof side, nothing new was proved: E adds nothing to Vampire, Prover9 and Twee here. Lemmas (the multi-variable consequences of a premise) or Waldmeister remain to be tried.
- Run E and the special magmas on the unconfirmed implications of the exploration over all ETP classes, and consider adding them to the Vampire expert.

8 Tools

- `eprover_batch.py`: poses the unproved laws of an exploration's unconfirmed implications and open questions to E, one law at a time, up to duality.
- `special_magmas.py RESULTS_DIR --equations FILE -o TABLES`: searches linear and translation-invariant magmas for counterexamples, checks each on its table, and writes the refuting tables for `--seed-magmas` or `--context-magmas`; tested in `tests/test_special_magmas.py`.

Both are in `explorations/equational_theories` and not yet committed.